Aarhus University

Computer Science

Numerical Linear Algebra

Exam

June 6, 2026

Problem 1

Given the basis E:

$$E: \ p_0(x) = 1 + x, \quad p_1(x) = 1 - x, \quad p_2(x) = x + x^2$$

How many of each p_i is needed to make $q(x) = 3 - 2x + x^2$, which is represented by a coordinate vector $[q]_E$.

This can also be explained by definition 19.6 and equation 19.1 [1] The answer is (B) $\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$

which amounts to:

$$0(1 + x) + 3(1 - x) + x - x^2 = 3 - 2x + x^2$$

So $[q]_E = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$ Answer (B)

Problem 2

For $V = \mathbb{R}^{3\times3}$ being the vector space for all real 3×3 matrices.
it is given in the start of chapter 25 that for $A^T = A$ then A is real symmetricthis means (B) hold as being in the subset of V
The answer is (B)

Problem 3

(a)

The shape can be determined by inserting an if statement for when $i = 3$
Note that the code is but a snippet, hence the import statement is ommited, and the final code is in the appendix, this will be the case for all code snippets.

```
    def mgs(a):
    q = np.copy(a)
    r = np.zeros((8, 8))
    for i in range(8):
        r[i, i] = np.linalg.norm(q[:, i])
        q[:, i] /= r[i, i]
        r[[i], i+1:] = q[:, [i]].T @ q[:, i+1:]

        if(i == 3 ):
            print((q[:, [i]] @ r[[i], i+1:]).shape)

        q[:, i+1:] -= q[:, [i]] @ r[[i], i+1:]
```

```
    return q, r

u = np.ones((120,8))
mgs(u)
```

The output is (120.8) which is the shape of $q[:, [i]]@r[[i], i+1:]$ when $i = 3$

(b)

The matrices are:

$$(q[:,\ [3]].T)^{1\times 120} \times (q[:,\ 3+1:])^{120\times 4}$$

so the resulting equation for calculating flops for matrix product is $2mkn = 2(1)(120)(4) = 960$
So the total amount of flaps used to compute the product is 960 flops.
The formula for flops can be found in chapter 5 section 5.1 [1]

Problem 4

the most important note here is the relation between the eigenvalues for A

$$-3, \quad 1, \quad 4, \quad 7.$$

and S

$$16, \quad 0, \quad 9, \quad 36.$$

The pattern is each eigenvalue in A has been subtracted by one and multiplied by itself which results in all positive eigenvalues.
this means the answer can't be (B) $S(A^2 - I_4)S^{-1}$ since it raises the power before subtracting with I_4.
If one uses theorem 7.2 [1] then $SS^{-1} = I_4$ and in answer (C) $(SAS^{-1} - i_4)^2$ this becomes $(AI_4 - I_4)^2$ where each diagonal in A is multiplied and subtracted by one before beign taken to the power of 2
In conclusion the answer is (C)

Problem 5

Given the vectors $w_0 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $w_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$ and $w_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

(a)

We form it's column matrix $W = [w_0|w_1|w_2]$

$$W = \begin{bmatrix} 1.00 & 1.00 & 0.00 \\ 0.00 & 0.00 & 1.00 \\ 1.00 & -1.00 & 0.00 \\ 0.00 & 0.00 & 1.00 \end{bmatrix}$$

then creating the gram-matrix from lemma 9.9 [1]

```
mat = np.array([[1,1,0],
                [0,0,1],
                [1,-1,0],
                [0,0,1]])
gram = mat.T @ mat
```

$$G = \begin{bmatrix} 2.00 & 0.00 & 0.00 \\ 0.00 & 2.00 & 0.00 \\ 0.00 & 0.00 & 2.00 \end{bmatrix}$$

Which according to lemma 9.9 if it's diagonal then it is orthogonal. and G is diagonal since all entries except the diagonal is zero.
Hence the collection w_0, w_1, w_2 is orthogonal

(b)

Problem 6

(a)

The system $Ax = y$ can be represented as

$$\begin{bmatrix} 1.0 & 0.0 & \cos(0) \\ 1.0 & 0.5 & \cos(0.5) \\ 1.0 & 1.0 & \cos(1) \\ 1.0 & 1.5 & \cos(1.5) \\ 1.0 & 2.0 & \cos(2.0) \\ 1.0 & 2.5 & \cos(2.5) \\ 1.0 & 3.0 & \cos(3.0) \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0.0 \\ 0.5 \\ 1.0 \\ 1.5 \\ 2.0 \\ 2.5 \\ 3.0 \end{bmatrix}$$

The dimension of $A = 3$ and $x = 3 \times 1$ and $y = 7 \times 1$

(b)

The function(s) used can be found in the appendix. It's a thin QR decompositon that uses the improve gram-schmidt and back substitution.

```
c = np.cos
t = np.array([0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0])
y = np.array([3.05, 2.48, 1.62, 0.63, -0.17, -0.56, -0.48])

A = np.array([[1, t[0], c(t[0])],
              [1, t[1], c(t[1])],
              [1, t[2], c(t[2])],
              [1, t[3], c(t[3])],
              [1, t[4], c(t[4])],
              [1, t[5], c(t[5])],
              [1, t[6], c(t[6])]])
x_solution = solve_least_squares_qr(A, y)

print(x_solution)
```

The least-squared solution for the coefficients in x are

$$\begin{bmatrix} 1.4188 \\ -0.3553 \\ 1.3114 \end{bmatrix}$$

Note that i've rounded the last decimal

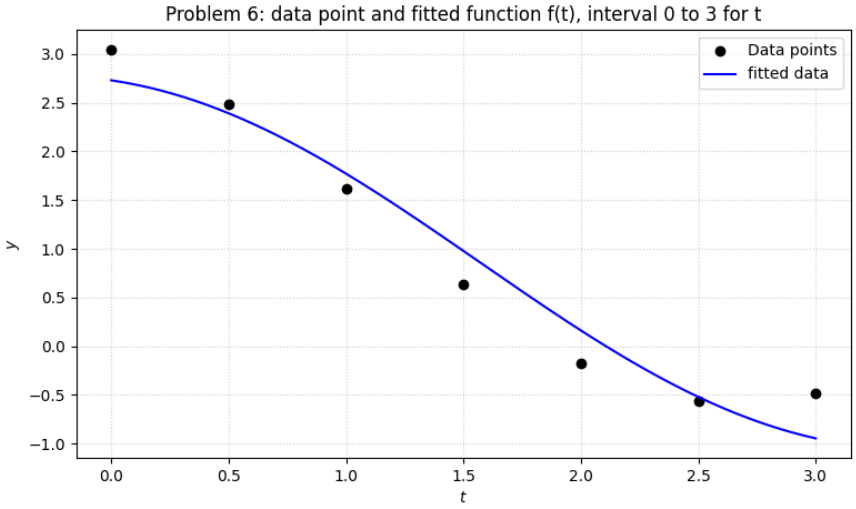
(c)

Using np.linspace and the function $f(t)$ with the coeeficients from (b), we fit it to the data points on the interval $0 \leq t \leq 3$
The code is a continuation of the previosu snippet from (b)

```
t_interval = np.linspace(0, 3 , 500)

yy_interval = x_solution[0,0]  + x_solution[1,0]*t_interval + x_solution
    [2,0]*c(t_interval)

plt.figure(figsize=(9, 5))
plt.plot(t, y, 'o', color='black', label='Data points')
plt.plot(t_interval, yy_interval, '-', color='blue', label=f'fitted data')
plt.xlabel('$t$')
plt.ylabel('$y$')
plt.title(f'Problem 6: data point and fitted function f(t), interval 0 to
    3 for t')
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend()
plt.show()
```



(d)

From chapter 16.1 [1] the residual vector is described aswell as the norm.
The residual vector is $r = b - Ax$:

$$r = \begin{bmatrix} 0.3198 \\ 0.0880 \\ -0,1520 \\ -0.3486 \\ -0.3324 \\ -0.0399 \\ 0.4653 \end{bmatrix}$$

and it's norm

$$||r||_2 = 0.7637$$

The following is the code used for this
Again, a continuation of the previous snippet.

```
yy_true = x_solution[0,0]  + x_solution[1,0]*t + x_solution[2,0]*c(t)

residual = y - yy_true
residual_norm = np.linalg.norm(residual)
deviation = np.abs(y - yy_true)

max_idt = np.argmax(deviation)
t_max_fail = t_interval[max_idt]
max_fail_val = deviation[max_idt]
```

The biggest deviation in the interval [0, 3] are 0.4653 and found with t = 0.0361
Which is the essentially the starting point.

Problem 7

Appendix

```
import numpy as np
import matplotlib.pyplot as plt
import sympy as sp

def back_subs(r, c):
    _, n = r.shape
    x = np.empty((n, 1))
    for i in reversed(range(n)):
        x[i] = (c[i] - r[[i], i+1:] @ x[i+1:]) / r[i, i]
return x


def forbedret_gram_schmidt(A):

    _, n = A.shape
    Q = A.copy().astype(float)
    R = np.zeros((n, n))
    for i in range(n):
        R[i, i] = np.linalg.norm(Q[:, i])
        Q[:, i] /= R[i, i]
        for j in range(i + 1, n):
            R[i, j] = np.vdot(Q[:, i], Q[:, j])
            Q[:, j] -= R[i, j] * Q[:, i]
    return Q, R

def solve_least_squares_qr(A, b):
    Q, R = forbedret_gram_schmidt(A)
    c = Q.T @ b
    return back_subs(R, c)


## Problem 3 code
def mgs(a):
    q = np.copy(a)
    r = np.zeros((8, 8))
    for i in range(8):
        r[i, i] = np.linalg.norm(q[:, i])
        q[:, i] /= r[i, i]
        r[[i], i+1:] = q[:, [i]].T @ q[:, i+1:]

        if(i == 3 ):
            print((q[:, [i]] @ r[[i], i+1:]).shape)

        q[:, i+1:] -= q[:, [i]] @ r[[i], i+1:]
    return q, r

u = np.ones((120,8))
mgs(u)
###############
```

```


### Problem 5 code
mat = np.array([[1,1,0],
                [0,0,1],
                [1,-1,0],
                [0,0,1]])
gram = mat.T @ mat


### Problem 6 code
c = np.cos
t = np.array([0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0])
y = np.array([3.05, 2.48, 1.62, 0.63, -0.17, -0.56, -0.48])

A = np.array([[1, t[0], c(t[0])],
              [1, t[1], c(t[1])],
              [1, t[2], c(t[2])],
              [1, t[3], c(t[3])],
              [1, t[4], c(t[4])],
              [1, t[5], c(t[5])],
              [1, t[6], c(t[6])]])
x_solution = solve_least_squares_qr(A, y)

print(x_solution)

t_interval = np.linspace(0, 3 , 500)

yy_interval = x_solution[0,0]  + x_solution[1,0]*t_interval + x_solution
    [2,0]*c(t_interval)

plt.figure(figsize=(9, 5))
plt.plot(t, y, 'o', color='black', label='Data points')
plt.plot(t_interval, yy_interval, '-', color='blue', label=f'fitted data')
plt.xlabel('$t$')
plt.ylabel('$y$')
plt.title(f'Problem 6: data point and fitted function f(t), interval 0 to
    3 for t')
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend()
plt.show()


yy_true = x_solution[0,0]  + x_solution[1,0]*t + x_solution[2,0]*c(t)

residual = y - yy_true
residual_norm = np.linalg.norm(residual)
deviation = np.abs(y - yy_true)

max_idt = np.argmax(deviation)
t_max_fail = t_interval[max_idt]
max_fail_val = deviation[max_idt]
```

Listing 1: Example Python code

References

[1] Paul D. Nelson. *Numerical Linear Algebra*. Institute for Mathematics, Aarhus University, 2026.